

## Python Programming Fundamentals

Python is a high-level language created by Guido van Rossum in 1991.

It emphasizes readability and simplicity. Python uses indentation instead of curly braces. Variables are dynamically typed.

## Python for Data Science

Key libraries include NumPy for numerical computing, Pandas for data manipulation, and Matplotlib for visualization. Scikit-learn provides machine learning tools. PyTorch and TensorFlow are deep learning frameworks. Jupyter Notebooks allow interactive development.

## Python Best Practices

Follow PEP 8 style guide. Functions should do one thing well.

Use meaningful variable names. Write docstrings for all functions.

Use virtual environments to isolate dependencies. Always pin dependencies in requirements.txt for reproducibility.